



uP – Proyecto Final

Smart Security Alarm System

uP Final Project

System and Computation Engineer

1. Introduction

From protecting our homes and businesses to safeguarding critical infrastructure, automated alarm systems form the first line of defense against unauthorized access and potential threats. These systems are no longer a luxury but a cornerstone of modern security protocols, providing peace of mind through constant, vigilant monitoring.

This project, the development of a Programmable **Multi-Zone Security System** using an Arduino MEGA, translates this vital real-world technology into an accessible and comprehensive learning experience. While commercial systems are often complex "black boxes," this endeavor deconstructs their core principles, demonstrating how simple, low-cost microcontrollers can be leveraged to create a robust and intelligent security framework.

In the industrial and commercial sectors, the concepts implemented here—zone-based monitoring, user authentication via PIN codes, and programmable delay timers—are ubiquitous. They allow for the flexible protection of large buildings, where different areas (e.g., server rooms, archives, manufacturing floors) require unique security levels and access permissions.

2. Outcomes

This project serves as a microcosm of these industrial systems. By integrating a diverse set of components—including an LCD for user interface, a keypad for access control, a relay for activating high-power alarms, and a network of LEDs and displays for system status—we move beyond simple component control to create a sophisticated state-based machine.

This project is uniquely valuable for System Engineers, as it uses the Arduino platform not as an end in itself, but as a powerful tool to abstract hardware complexities and focus on core systems engineering principles. By leveraging a pre-integrated developer kit, we intentionally minimize the focus on electronic design and circuit analysis, thereby amplifying the complexity and importance of the software architecture and system logic design: state logic, user interaction, protocol management, and architectural integrity.

The key learning outcomes include:

- **System Integration:** Mastering the orchestration of multiple input and output devices into a cohesive, reliable unit.



- State Machine Design: Implementing complex logical workflows (Disarmed, Arming, Armed, Triggered) that are fundamental to embedded systems programming.
- Human-Machine Interface (HMI): Designing an intuitive user experience with clear visual feedback, which is critical for any real-world product.
- Problem Decomposition: Breaking down a complex problem (security) into manageable, programmable subsystems (sensing, logic, alerting, user control).

Ultimately, this project is more than a simple assembly of parts; it is a functional prototype that embodies the principles of automation, control engineering, and user-centered design. It provides a tangible demonstration of how a handful of basic electronic components, guided by thoughtful programming, can be transformed into a system with significant practical utility, laying the groundwork for understanding the sophisticated security technologies that permeate our daily lives.

3. Alarm System – Basic Conditions & Design

3.1. System States

The alarm will operate in the following main states:

1. DISARMED: System is off and inactive.
2. ARMING: Countdown period to exit in the premises.
3. ARMED: System is active and monitoring sensors.
4. TRIGGERED: Alarm has been activated, sirens sounding.
5. ALARM_MEMORY: After alarm stops, show which sensor triggered it.

3.2. Core Components & Their Functions

1. Input Components (“Sensors” & Controls)
 - a. 4x4 Keypad: Enter PIN code to arm/disarm the system
 - b. DIP Switches (8):
 - i. Switch 1-4: Intrusion sensor zones
 - ii. Switch 6: Silent alarm (no siren) vs. audible
 - iii. Switch 7-8: Reserve for future features
 - c. Potentiometers:
 - i. Port1: Adjust arm/disarm delay timing
 - ii. Port2: Adjust alarm duration/siren length
 - d. Push buttons:
 - i. Button1: Panic button (instant alarm)
 - ii. Button2: Quick arm/disarm toggle
2. Output Components (Indicators & Alerts)
 - a. LCD Display: Show system status, messages, and prompts
 - i. System DISARMED

- ii. ARMING... Exit now! XXs
- iii. SYSTEM ARMED – Stay back!
- iv. ALARM! Zone X triggered

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- v. Enter PIN: ****
- b. 7-Segment Display (4 digit)
 - i. Countdown timer during arming/disarming
 - ii. Zone number when alarm is triggered
- c. LEDs (8):
 - i. LED1: Power indicator (always on)
 - ii. LED2: Armed status (steady when armed)
 - iii. LED3-6: Zone status (show which zones are active/violated)
 - iv. LED7: Alarm memory (flashing after alarm)
 - v. LED8: System fault/error
- d. Relé: Main alarm output – activates siren/strobe light

3.3. Basic Operational Logic

Arming Sequence:

- User enters valid PIN on keypad
- System shows: ARMING... Exit now!
- 7-segment displays countdown from 30 seconds (adjustable via Pot1)
- LEDs flash during countdown
- After countdown: System moves to ARMED state

Armed State:

- LCD shows: SYSTEM ARMED
- Armed LED (LED2) lights steadily
- System monitors "sensor inputs"

Alarm Triggering:

- Violation: When a "sensor" is triggered
- Siren: Relay activates for duration set by Pot2
- Display: LCD shows which zone was violated
- 7-segment: Shows zone number
- LEDs: Zone indicator LED flashes rapidly

Disarming:

- User enters valid PIN during alarm or when armed
- If during alarm: siren stops immediately
- System returns to DISARMED state
- Alarm memory LED remains flashing until reset

Consider to use serial monitor commands for testing

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